CSM
Spring 2026

EECS 16B
# Worksheet 3

**1. Complex Number Practice**

(a) Take $z_1$ and $z_2$ to have the following rectangular coordinates form:

$$z_1 = 2 + 6j$$
$$z_2 = 3 - 5j.$$

What is the real part of $z_1 + z_2$?

(b) What is the imaginary part of $z_1 + z_2$?

(c) How do we visualize the polar representation of complex numbers, $z = Ae^{j\theta}$, in the complex plane?

(d) Given $z = 7e^{j30^\circ}$, what is $|z|$?

(e) What is $\angle z$?

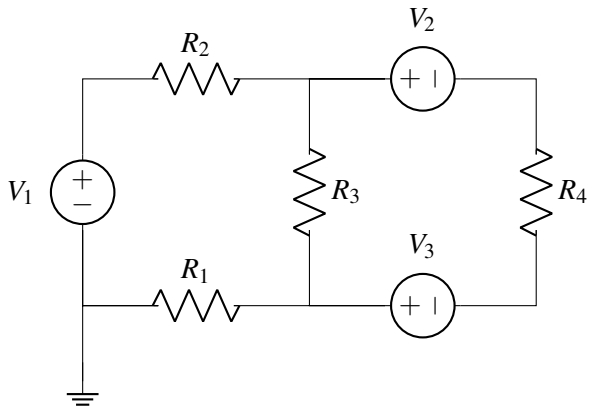
(f) Given $z = 5e^{j90^\circ}$, what is the real part of $z$?

(g) What is the imaginary part of $z$?

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## 2. Supernode

In this question, we will explore how to deal with multiple voltage sources when doing nodal analysis. Throughout the problem, assume that $R_1 = 1\Omega, R_2 = 2\Omega, R_3 = 4\Omega, R_4 = 4\Omega, V_1 = 1\text{V}, V_2 = 2\text{V}, V_3 = 4\text{V}$. While it is good practice to symbolically solve problems before substituting specific numbers, for superposition it is often easiest to substitute before superposing the solwers.




(a) Mark all the nodes. If you know the potential at the node, write down the value next to the node. If you don't know the value, then assign a variable for the potential.

(b) Mark current directions arbitrarily and corresponding polarities on each resistor. Note that if current goes from left to right, then the left side of the resistor is to be marked + and the right side must be marked -. This is the passive sign convention.

(c) Note that we define 4 nodes with unknown potentials. So we need 4 equations. Each of these nodes with unknown potential should give us one equation.
Write the equation for the first node with unknown potential.

(d) Write equations for all remaining unknown nodes. Solve for all the unknown potentials.

(e) Now we will explore solving the circuit using superposition. To recall the process, which 2 sources will be suppressed when looking at each of the 3 sources. Will the other 2 sources be treated as open circuits or short circuits?

(f) Draw the superposition circuit for $V_1$ and solve this circuit.

(g) Draw the superposition circuit for $V_2$ and write the equation that characterizes this circuit.

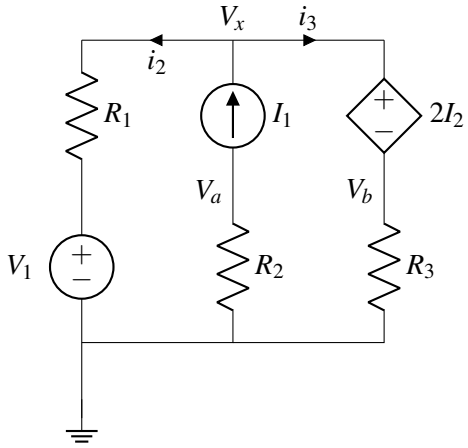
(h) Draw the superposition circuit for $V_3$ and write the equation that characterizes this circuit.

(i) Find the unknown potentials as you did in in part (d), but now do so using the superposition theorem.

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**3. It's a bird! It's a plane! No, it's superposition!**

(a) Solve the following circuit for $V_x$ using nodal analysis. Let $R_1 = 10\Omega$, $R_2 = 5\Omega$, $R_3 = 2\Omega$, $V_1 = 7V$, and $I_1 = 6A$.




(b) Now solve the same circuit for $V_x$ using superposition.

**Contributors:**

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